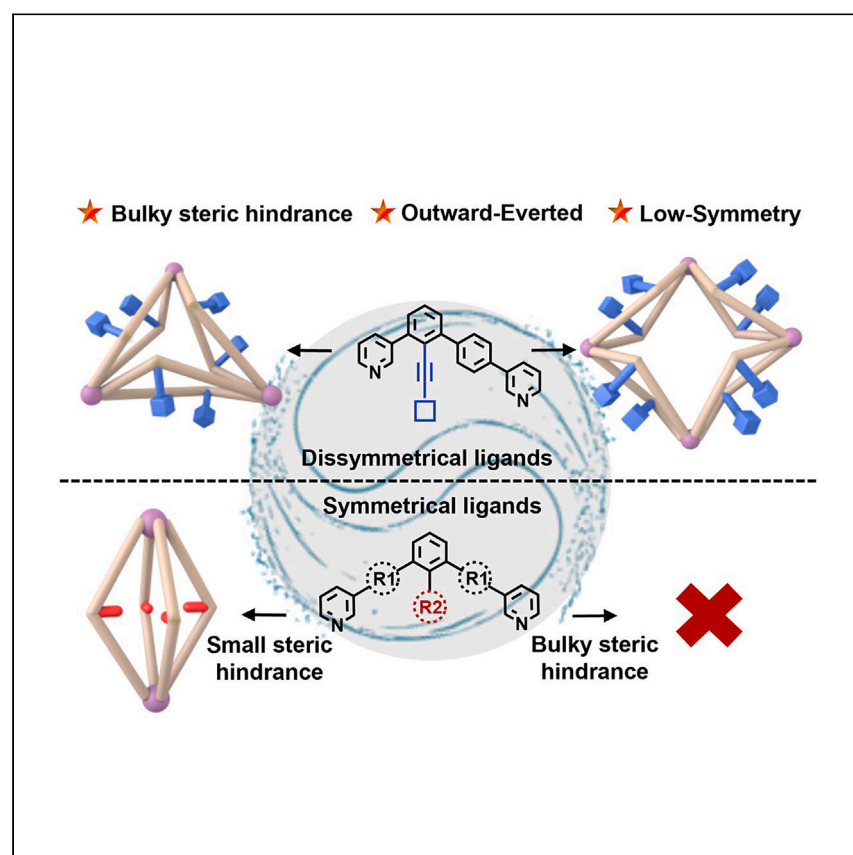


Article

Construction of outward-everted metal-organic cages induced by steric hindrance groups based on dissymmetrical ligands



Wang et al. strategically modify dissymmetrical bipyridine ligands with bulky steric hindrance groups and successfully assembled them with Pd(II) to form outward-everted hexameric and octameric cages. However, symmetrical ligands modified with bulky hindrance groups fail to form discrete supramolecular structures.

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Highlights

A strategy for constructing outward-everted metal-organic cages

The ligands in low-symmetry metal-organic cages possess diverse chemical environments

Comparison of dissymmetrical ligand and symmetrical ligand assembly

Yu et al., Cell Reports Physical Science 4, 101631

October 18, 2023 © 2023 The Author(s).

<https://doi.org/10.1016/j.xcrp.2023.101631>



Article

Construction of outward-everted metal-organic cages induced by steric hindrance groups based on dissymmetrical ligands

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SUMMARY

Steric hindrance groups hold a pivotal role in the self-assembly process of biomolecules. Building on this insight, a strategy for synthesizing outward-everted molecular cages by *endo*-modification of steric hindrance groups on dissymmetrical ligands is proposed in this work. Specifically, the dissymmetrical ligands have been *endo*-modified with five distinct bulky hindrance groups, and they are respectively assembled with Pd(II). In order to relieve steric crowding, all hindering groups are oriented outward from the cage, and the arrangement of the two dissymmetrical ligands between adjacent Pd(II) is opposite, indicating high flexibility in regulation. As a result, a low-symmetry hexameric or octameric cage with dissimilar chemical environments for the ligands can be assembled. This work provides valuable insights and has significant implications for the construction of new supramolecular systems.

INTRODUCTION

The process of self-assembly in biomolecules is an intricate and captivating phenomenon that plays a fundamental role in various biological processes.^{1,2} One crucial factor that exerts a substantial influence on the self-assembly of biomolecules is the steric hindrance group. The presence of bulky side groups or substituents gives rise to steric hindrance, which in turn affects the spatial arrangement and interactions of biomolecules. These steric effects can have notable implications on the conformational flexibility, stability, and functionality of biomolecules.^{3,4} A prime example of steric hindrance effects can be observed during the assembly of proteins, where the distinctive R-group side chains of different amino acids vary significantly in size and shape, ultimately shaping the overall structure and functionality of the protein.^{5,6} In the case of amino acids with an internal bulky R group, such as tryptophan,⁷ proline,⁸ tyrosine,⁹ etc., space congestion may arise within the protein due to the steric hindrance. This necessitates specific conformational changes in the protein structure to accommodate the dimensions of the R group. To emulate the intricacy and efficacy of natural substances, scientists employed self-assembly techniques to transform basic molecular precursors into complex supramolecular architectures.^{10–15} In particular, the use of metal-ligand coordination interactions has gained popularity in the construction of discrete supramolecular complexes with predetermined structures owing to their appropriate bond energy and strong directionality.^{16–29} As a result, the implementation of metal-coordinated self-assembly represents a promising approach to emulate the impact of steric hindrance groups observed in biomolecules.

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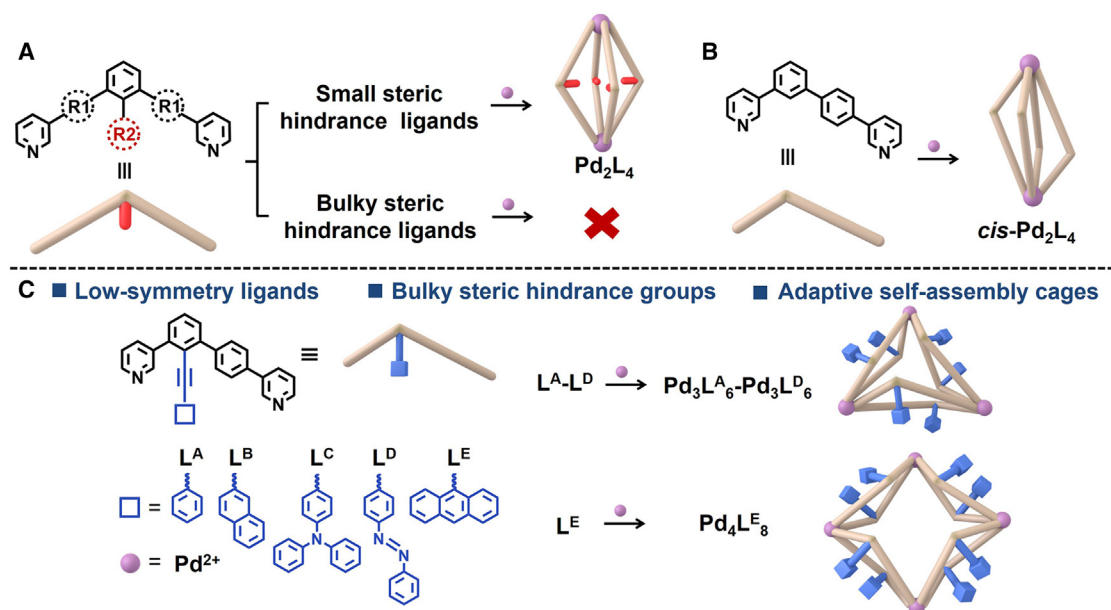
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<https://doi.org/10.1016/j.xcrp.2023.101631>





Scheme 1. Self-assembly of symmetrical and dissymmetrical ligands endo-modified by steric hindrance groups with Pd(II)

(A) A common strategy for constructing endo-functionalized metal-organic cages using symmetrical ligands.

(B) Construction of the single-configuration *cis*- Pd_2L_4 cages via the dissymmetrical ligand.

(C) In this study, outward-everted metal-organic cages were constructed by endo-modification of bulky steric hindrance groups in dissymmetrical ligands.

Over the past few decades, the cooperative assembly of dipyrindyl organic ligands and transition metal Pd/Pt(II) to construct metal-organic cages has been one of the focuses in the supramolecular field.^{30–40} Fujita's research group has demonstrated pioneering advancements in the field of metal-organic cage structures.^{41–43} Subsequent research efforts by groups such as Clever,^{44,45} Crowley,^{46,47} Yoshizawa,^{48,49} and Severin^{50,51} have also yielded numerous innovative cage structures with versatile applications across various fields. A noteworthy aspect of these structures is that the organic ligands are highly modifiable, allowing researchers to create metal-organic cages by modifying the organic ligands with various functional groups.^{52–58} Typically, endo-functionalized classical Pd_2L_4 cages are formed using symmetrical meta-pyridine ligands and Pd(II) coordination (Scheme 1A). However, the highly symmetrical structure of the final assembly requires internal modifying groups with small steric volume to accommodate four modifying groups in the inner cavity of the molecular cage.^{59–63} Unfortunately, this limitation restricts the assembly of bulky functional groups within organic ligands. In a previous investigation, we successfully employed the flexible tuning ability of dissymmetrical ligands to direct the synthesis of single-configuration *cis*- Pd_2L_4 cages (Scheme 1B).⁶⁴ Drawing from this, we hypothesize that the introduction of a bulky steric hindrance functional group into the dissymmetrical ligand facilitates complementary coordination through the length disparity on each side of the molecule during assembly, thereby ameliorating the issue of spatial crowding. By adopting this approach, we are able to simulate the impact of steric hindering groups on biomolecules while also simultaneously addressing the challenge associated with the modification of bulky steric hindrance groups internally during the assembly of organic ligands. Furthermore, the assembly of dissymmetrical ligands with Pd(II) can be employed to construct metal-organic cages with reduced symmetry.^{31,65,66} These assemblies exhibit distinctive inner cavities that have the potential to engender unique functional attributes, such as recognition of specific molecules⁴⁶ and chirality.⁶⁷

Herein, we present a strategy based on dissymmetrical ligands for constructing outward-everted metal-organic cages (Scheme 1C). This approach involves the *endo*-modification of bulky steric hindrance groups onto dissymmetrical ligands, resulting in outward flipping of ligands and the formation of low-symmetry metal-organic cages. Specifically, we designed and synthesized five types of dissymmetrical ligands, namely L^A – L^E , each containing different steric hindrance groups such as benzene, naphthalene, triphenylamine, azobenzene, and anthracene, linked by alkyne bonds. By assembling these dissymmetrical ligands with Pd(II), we successfully obtained outward-everted low-symmetry hexameric cages $Pd_3L^A_6$ – $Pd_3L^D_6$ and octameric cage $Pd_4L^E_8$. Conversely, when attempting to assemble the corresponding symmetrical ligands modified with steric hindrance groups, all failed to form discrete supramolecular structures, except ligand L^F with minimum steric hindrance. In these low-symmetry molecular cages, in order to alleviate the space crowding brought by the large hindering groups, the arrangement of two dissymmetrical ligands between adjacent Pd(II) is opposite, and all the hindering groups point to outside the cage.

RESULTS AND DISCUSSION

Assembly of dissymmetrical ligand L^A

The synthesis and characterization of dissymmetrical ligands L^A – L^E are shown in the supplemental information (Schemes S1–S5, Figures S1–S4, S9–S12, S17–S19, S25–S27, S34–S37, and S64–S68). In this study, we first constructed a Pd(II) complex of L^A by mixing L^A and $Pd(CH_3CN)_4(BF_4)_2$ at a precise stoichiometric ratio of 2:1 in DMSO- d_6 . Compared with the 1H nuclear magnetic resonance (NMR) spectrum of L^A (Figure 1A), its Pd(II) complex (Figure 1B) showed an increased number of proton signals after coordination with Pd(II). Integration of the proton signal showed that the ligand L^A in this complex was present in three different chemical environments with a 1:1:1 ratio (Figure S5). Notably, a downfield shift of the hydrogen signals H^m , H^a , H^b , and H^l on pyridine groups was observed upon coordination of L^A with Pd(II), which was attributed to a reduction in electron cloud density after coordination. Furthermore, proton signals H^p , H^q , and H^n originating from the hindered alkyne phenyl group transformed into three distinct peaks and exhibited upfield shifts after coordination. The 2D diffusion ordered spectroscopy (DOSY) NMR of the complex revealed only one band with $\log D = -10.05$ (Figure S7), indicating that this complex is a single-component product. The experimental hydrodynamic radius (r_H) value calculated via the Stokes-Einstein equation was 1.09 nm. The molecular composition of the obtained complex was determined by electrospray ionization mass spectrometry (ESI-MS) (Figures 1C, S74, and S75), which detected a series of peaks with a continuum of charge states corresponding to the loss of different amounts of BF_4^- counterions. After deconvolution, the molecular weight of the L^A assembly was calculated to be 3,288.7 Da, corresponding to six ligands, three Pd(II) ions, and six BF_4^- . Importantly, the analysis revealed that the assembly of L^A and Pd(II) did not form a common tetramer structure, Pd_2L_4 , but rather formed a hexameric complex, $Pd_3L^A_6$, under the effect of the steric hindrance.

By using NMR and MS, we established the formation of a hexameric complex, $Pd_3L^A_6$, with three distinct chemical environments, which drives us to further determine the exact configuration of $Pd_3L^A_6$ through crystal characterization. After diffusing ethyl acetate vapor into $Pd_3L^A_6$ in DMSO for 3 weeks, we obtained testable crystals and collected single-crystal data by synchrotron radiation. Surprisingly, our crystal structure analysis revealed that the assembly was not the

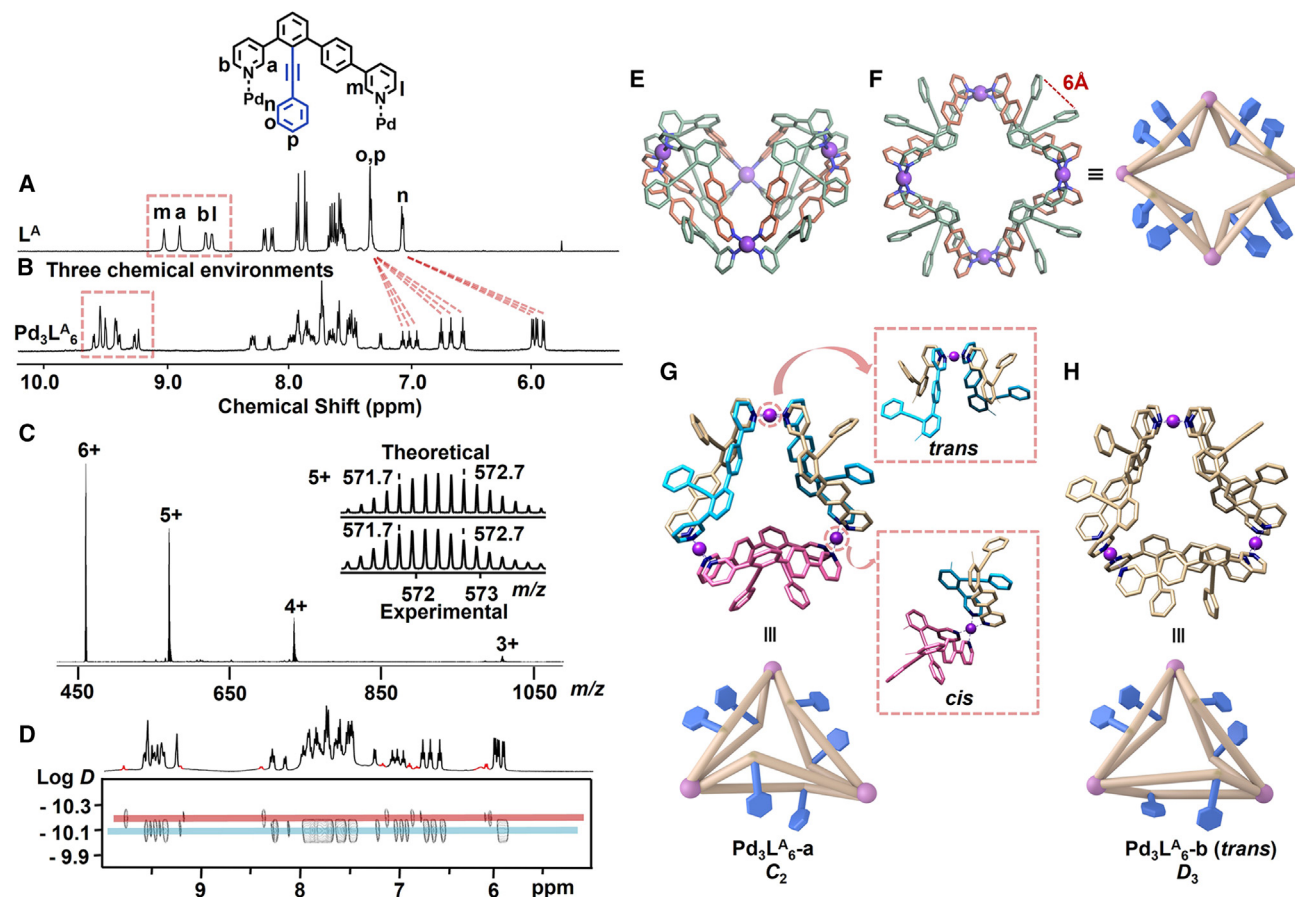


Figure 1. Characterization of hexamer Pd_3L_6 and octamer Pd_4L_8

- (A) ^1H NMR spectrum (600 MHz, 300 K) of ligand L^{A} at 5 mg/mL in $\text{DMSO}-d_6$.
 (B) ^1H NMR spectrum (600 MHz, 300 K) of Pd_3L_6 at 5 mg/mL in $\text{DMSO}-d_6$.
 (C) ESI-MS spectrum of Pd_3L_6 .
 (D) 2D DOSY NMR spectrum of a mixture of Pd_3L_6 and Pd_4L_8 at 40 mg/mL in $\text{DMSO}-d_6$.
 (E) The front view of the crystal structure of Pd_4L_8 (hydrogen atoms are omitted for clarity).
 (F) The top view of the crystal structure of Pd_4L_8 (hydrogen atoms are omitted for clarity).
 (G) The structural simulation of Pd_3L_6 -a.
 (H) The structural simulation of Pd_3L_6 -b.

solution-state hexamer Pd_3L_6 but instead the octamer Pd_4L_8 with a larger structure (Figures 1E and 1F). Interestingly, to avoid space crowding during the coordination assembly, the alkyne benzene hindering groups flipped out of the cage, and the two dissymmetrical ligands between adjacent Pd(II) showed up and down staggered, long and short complementary arrangements. Additionally, all ligands in Pd_4L_8 were *cis*-linked, the structure displayed D_{2d} symmetry, and the molecular size was 2.54 nm with a 6 Å spacing between the benzene groups of two adjacent hindering groups (Table S1; Figures S91 and S92). We repeated the crystal growth and testing process and obtained the same results, leading us to speculate that this was due to the structural transformation of Pd_3L_6 to Pd_4L_8 as the concentration increased during the gradual precipitation of crystals. To verify this conjecture, we increased the concentration of Pd_3L_6 and observed a new set of proton peaks in the ^1H NMR spectrum at a concentration of 40 mg/mL. Next, the 2D DOSY spectrum showed two bands, corresponding to the main product peak ($\log D = -10.11$) and the new proton peak ($\log D = -10.19$) (Figure 1D),

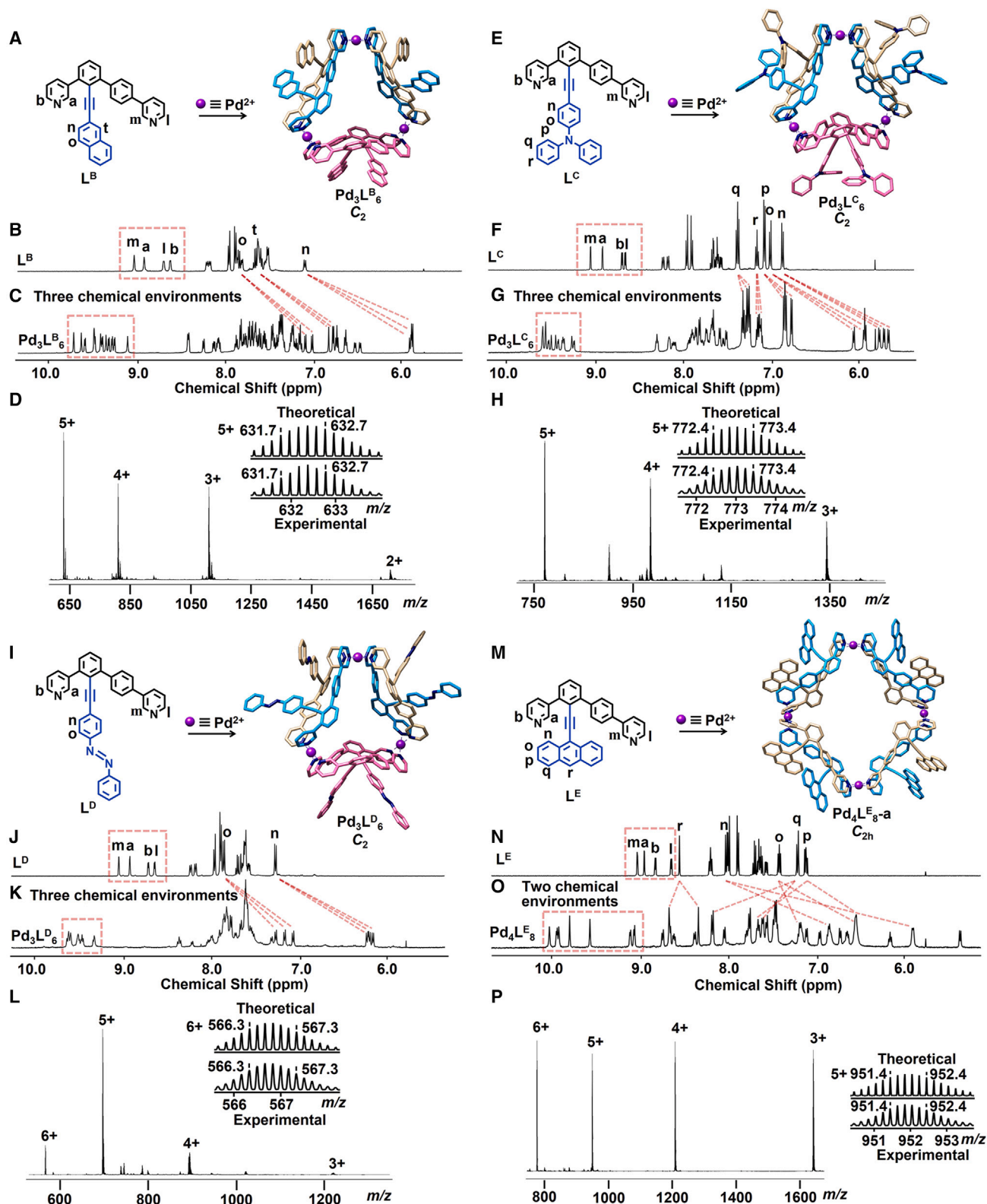


Figure 2. Characterization of hexamer $\text{Pd}_3\text{L}^{\text{B}}_6$, $\text{Pd}_3\text{L}^{\text{C}}_6$, $\text{Pd}_3\text{L}^{\text{D}}_6$, and octamer $\text{Pd}_4\text{L}^{\text{E}}_8$

(A, E, I, and M) Self-assembly of (A) $\text{Pd}_3\text{L}^{\text{B}}_6$, (E) $\text{Pd}_3\text{L}^{\text{C}}_6$, (I) $\text{Pd}_3\text{L}^{\text{D}}_6$, and (M) $\text{Pd}_4\text{L}^{\text{E}}_8$.
(B, C, F, G, J, K, N, and O) ^1H NMR spectra (600 MHz, 300 K) of (B) L^{B} and (C) $\text{Pd}_3\text{L}^{\text{B}}_6$, (F) L^{C} and (G) $\text{Pd}_3\text{L}^{\text{C}}_6$, (J) L^{D} and (K) $\text{Pd}_3\text{L}^{\text{D}}_6$, and (N) L^{E} and (O) $\text{Pd}_4\text{L}^{\text{E}}_8$ at 5 mg/mL in $\text{DMSO}-d_6$.
(D, H, L, and P) ESI-MS spectra of (D) $\text{Pd}_3\text{L}^{\text{B}}_6$, (H) $\text{Pd}_3\text{L}^{\text{C}}_6$, (L) $\text{Pd}_3\text{L}^{\text{D}}_6$, and (P) $\text{Pd}_4\text{L}^{\text{E}}_8$.

indicating that the additional compound was the octamer $\text{Pd}_4\text{L}^{\text{A}}_8$ with a larger structure.

Through crystal analysis, we determined that in the complex $\text{Pd}_4\text{L}^{\text{A}}_8$, the arrangement of the two dissymmetrical ligands between adjacent Pd(II) was opposite, and outward eversion of the ligands caused alkyne benzene hindering groups to point out of the cage. Therefore, we hypothesize that the dissymmetrical ligands in the hexameric cage $\text{Pd}_3\text{L}^{\text{A}}_6$ were also arranged in the same way to relieve the steric crowding caused by the bulky hindrance groups. Based on this, we simulated the only two configurations in which the ligands could meet the above arrangement, $\text{Pd}_3\text{L}^{\text{A}}_6\text{-a}$ and $\text{Pd}_3\text{L}^{\text{A}}_6\text{-b}$. In $\text{Pd}_3\text{L}^{\text{A}}_6\text{-a}$, one Pd(II) center's linker motif is in a *trans*-configuration, while the linker motifs of the other two Pd(II) centers are in a *cis*-configuration (Figure 1G; Video S1). The structure had C_2 symmetry with low symmetry, resulting in ligands in the $\text{Pd}_3\text{L}^{\text{A}}_6\text{-a}$ assembly having three chemical environments. Figure 1G depicts the differentiation of ligands in varying chemical environments through three diverse colors. The ratio of the ligands in different chemical environments was determined to be 1:1:1, consistent with the one-dimensional NMR experimental result. In addition, the molecular diameter of $\text{Pd}_3\text{L}^{\text{A}}_6\text{-a}$ was determined to be 2.03 nm, which agrees well with the 2D DOSY experimental results. In $\text{Pd}_3\text{L}^{\text{A}}_6\text{-b}$, all the ligands were connected in *trans*, and the molecular structure was D_3 symmetric, resulting in only one chemical environment of ligands in the assembly, which did not match our NMR experimental results (Figure 1H). Thus, based on these findings, we assign the hexameric complex of L^{A} and Pd(II) as $\text{Pd}_3\text{L}^{\text{A}}_6\text{-a}$. Notably, the low symmetry of the assembly of $\text{Pd}_3\text{L}^{\text{A}}_6\text{-a}$ results in a pair of enantiomers (Figure S93).

Assembly of dissymmetrical ligands L^{B} – L^{E}

A outward-everted hexameric cage $\text{Pd}_3\text{L}^{\text{A}}_6$ was successfully constructed by using alkyne benzene as a steric hindrance group on the dissymmetrical ligand. Indeed, the construction of the low-symmetry molecular cage using a single ligand and achieving diverse chemical environments is a rare occurrence.^{68,69} This presents a distinctive opportunity to gain significant insights into the self-assembly process, thereby enhancing our comprehension of the formation and properties of intricate systems.⁷⁰ To investigate the generality of this construction strategy, alkyne-linked naphthalene, triphenylamine, azobenzene, and anthracene with larger steric hindrance were used to obtain dissymmetrical ligands L^{B} , L^{C} , L^{D} , and L^{E} . The Pd(II) complexes based on these four ligands were obtained in high yields by using the same assembly method as $\text{Pd}_3\text{L}^{\text{A}}_6$. Notably, the Pd(II) complexes of L^{B} , L^{C} , and L^{D} exhibited similar NMR patterns to that of $\text{Pd}_3\text{L}^{\text{A}}_6$ (Figures 2C, 2G, and 2K). Upon coordination with Pd(II), the proton signals of the assemblies split into three distinct sets of signals with identical integrals (Figures S13, S20, and S28), indicative of the existence of three chemical environments for the ligands within these assemblies. Furthermore, the proton signals on the pyridine moiety experienced a downfield shift, while the proton signals on the steric group underwent varying degrees of upfield shifts upon coordination. The 2D DOSY NMR of the three assemblies showed only one band (Figures S15, S23, and S30). The log *D* values of the complexes of L^{B} , L^{C} , and L^{D} with Pd(II) were –10.07, –10.10, and –10.19, respectively, and the experimental r_{H} calculated by the Stokes-Einstein equation corresponded to the

radius of the expected hexameric structures (1.15, 1.24, and 1.54 nm, respectively). The molecular composition of the three assemblies was also further characterized by ESI-MS (Figures 2D, 2H, 2L, and S76–S81). After deconvolution, the molecular weights of these complexes constructed by L^B , L^C , and L^D with Pd(II) are 3,588.8, 4,291.14, and 3,912.93 Da, respectively, corresponding to six ligands, three Pd(II) ions, and six BF_4^- . This indicated that these three assemblies are hexamer $Pd_3L^B_6$, $Pd_3L^C_6$, and $Pd_3L^D_6$, respectively. Based on these evidence, we infer that the structural configuration of the three assemblies is analogous to that of $Pd_3L^A_6$, with steric groups oriented outward from the cage. Additionally, the two dissymmetrical ligands between the Pd(II) atoms adopt a staggered up and down configuration, which helps to alleviate the steric crowding caused by the large steric hindrance groups. We have simulated the structures of $Pd_3L^B_6$, $Pd_3L^C_6$, and $Pd_3L^D_6$ in Figures 2A, 2E, and 2I, corresponding to Videos S2, S3, and S4, respectively. The resulting molecular structures are all C_2 symmetric, and their diameters (2.2, 2.76, and 2.81 nm) are basically consistent with those obtained from DOSY experiments. Furthermore, the low symmetry of the structures results in the existence of a pair of enantiomers in all three assemblies (Figures S94–S96). As a matter of fact, our investigation also delved into the interconversion of the *trans*- and *cis*-configurations of azobenzene in L^D and $Pd_3L^D_6$. Elaborate experimental findings pertaining to this aspect can be found in the supplemental information.

The 1H NMR spectra of L^E and its Pd(II) complex are presented in Figures 2N and 2O, respectively. Compared with the previous four ligands, the assembly based on L^E only showed two distinct chemical environments, and the ratio in these two environments was determined via NMR integration to be 1:1 (Figure S38). Upon coordination with Pd(II), the proton signals H^m , H^a , H^b , and H^l on the pyridine groups exhibited a downfield shift due to metal coordination, while the steric hindrance group 9-ethyne anthracene proton signal experienced a different chemical shift. The molecular composition of the L^E complex was determined via ESI-MS to consist of eight ligands, four Pd(II) ions, and eight BF_4^- (Figures 2P, S83, and S84), indicating that the assembly of L^E and Pd(II) under the influence of the 9-ethyne anthracene steric group resulted in the formation of an octamer complex, $Pd_4L^E_8$, rather than a hexamer. As additional evidence, 2D DOSY NMR was used to further confirm the formation of the octamer. In Figure S40, only one band was observed ($\log D = -10.10$), and the experimental r_H calculated via the Stokes-Einstein equation was 1.24 nm, agreeing well with the corresponding octamer structure. This is attributed to the steric hindrance within L^E caused by the 9-anthracene with a larger lateral distance. During the assembly process, this adaptive self-assembly way of L^E into a larger octameric structure is beneficial for reducing the impact of steric hindrance. Drawing from the crystal structure of $Pd_4L^A_8$ and previous studies, we proposed that the 9-ethyne anthracene hindering groups in $Pd_4L^E_8$ are still oriented toward the outside of the cage and that the arrangement of the dissymmetrical ligands between the two Pd(II) atoms is reversed. Based on this assumption, we simulated only four configurations that could satisfy this arrangement, $Pd_4L^{E_8-a}$, $Pd_4L^{E_8-b}$, $Pd_4L^{E_8-c}$, and $Pd_4L^{E_8-d}$ (Figure 2M; Video S5; Figure S97). Among these, only the $Pd_4L^{E_8-a}$ structure exhibited a C_{2h} symmetry and possessed two chemical environments for ligands, with a 1:1 ratio consistent with our NMR results. $Pd_4L^{E_8-d}$ had a lower C_2 symmetry and four chemical environments for ligands, but they did not match our NMR results. Both $Pd_4L^{E_8-b}$ and $Pd_4L^{E_8-c}$ had high symmetry and only one chemical environment for ligands in the assembly, which was also inconsistent with our NMR results. Therefore, we conclude that the octameric complex of L^E with Pd(II) is $Pd_4L^{E_8-a}$. It is important to highlight that, following the characterization of the assembly of all dissymmetrical ligands, we conducted additional experiments. Firstly,

Substituent Assembly of Pd(II)					
	L ^A ↓ Pd ₃ L ^A ₆	L ^B ↓ Pd ₃ L ^B ₆	L ^C ↓ Pd ₃ L ^C ₆	L ^D ↓ Pd ₃ L ^D ₆	L ^E ↓ Pd ₄ L ^E ₈
	L ^F ↓ Pd ₃ L ^F ₆	L ^G ↓ ×	L ^H ↓ ×	L ^I ↓ ×	L ^J ↓ ×

Figure 3. Effect of symmetry on assembly

Comparative assembly of dissymmetrical and symmetrical ligands *endo*-modified by bulky steric hindrance groups with Pd(II).

we verified that the presence of a moderate excess of Pd(II) salt did not exert any discernible influence on the resulting assembly product (Figures S62 and S63). Secondly, we confirmed that the co-assembly of two dissymmetrical ligands with Pd(II) led to the formation of a statistical mixture of the two ligands (Figures S87 and S88).

Assembly of symmetrical ligands L^F–L^J

To investigate the feasibility of the steric hindrance group strategy for constructing symmetrical ligand assemblies, we synthesized L^F–L^J corresponding to L^A–L^E and investigated their assembly with Pd(II) (Figure 3). Notably, only the assembly of L^F modified with the smaller hindered group alkyne benzene with Pd(II) yielded a discrete hexameric molecular cage, Pd₃L^F₆. Conversely, L^G–L^J featuring larger steric hindrance groups failed to assemble into discrete complexes with Pd(II) due to the incapability of mitigating excessive steric crowding during the assembly process.

The synthesis and characterization of the symmetrical ligands L^F–L^J are presented in the supplemental information (Schemes S6 and S7; Figures S42–S44, S49–S60, and S69–S73). Upon coordination of the ligand L^F with Pd(II), a single set of main product peaks was observed in ¹H NMR, indicating the formation of a highly symmetrical assembly (Figure 4A). ESI-MS confirmed the molecular weight of the complex to be 3,744.89 Da, corresponding to Pd₃L^F₆ with six ligands L^F, three Pd(II) ions, and six BF₄[−] (Figures S85 and S86). Density functional theory (DFT) calculations revealed that Pd₃L^F₆ maintains the configuration with the steric hindrance groups being exposed outwardly (Figure S98). The ¹H NMR spectra of Pd(II) complexes of L^G, L^H, L^I, and L^J are depicted in Figures 4B–4E, respectively. The NMR proton signals observed for these complexes exhibit a complex and disordered pattern of peaks, which indicated that the assembly of these four symmetrical ligands does not form a single discrete complex. In addition, ESI-MS characterization was performed on the L^G–L^J and Pd(II) complex, and MS information regarding the discrete molecular cages was also not detected (Figure S89). This failure can be attributed to the larger steric hindrance groups in the symmetrical ligands. In contrast, dissymmetrical ligands with similar steric hindrance groups can create discrete metal-organic cages,

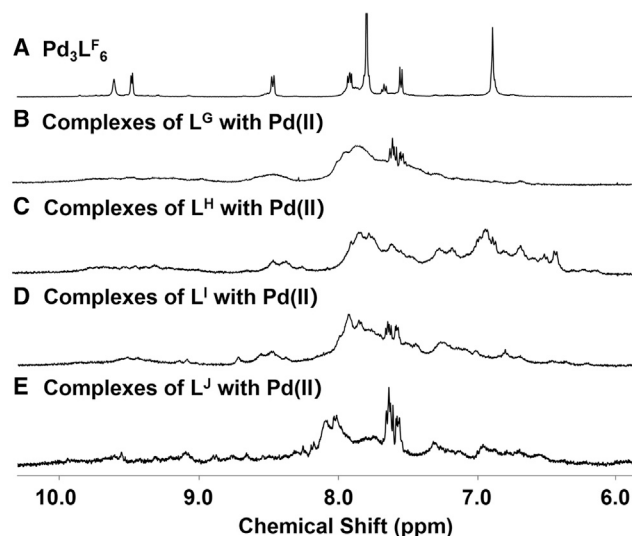


Figure 4. ^1H NMR spectra of symmetrical ligand assembly products

^1H NMR spectra of (500 MHz, $\text{DMSO}-d_6$, 300 K) (A) Pd_3L_6 and complexes of Pd(II) with ligands (B) L^G , (C) L^H , (D) L^I , and (E) L^J .

due to dissymmetrical ligands can cause steric hindrance to be arranged in a dislocated manner, thereby reducing the impact and resulting in greater flexibility for tuning assembly.

In summary, a strategy for the synthesis of outward-everted molecular cages is proposed by *endo*-modification of bulky hindrance groups onto dissymmetrical ligands. Five different bulky steric hindrance groups are introduced onto dissymmetrical ligands, which are subsequently assembled with Pd(II) to form hexameric or octameric cages with low symmetry. During the assembly process, the steric hindrance groups are strategically positioned outside of the cage to alleviate space congestion, while the two dissymmetrical ligands between adjacent Pd(II) are arranged in an opposite manner. This arrangement illustrates the flexible coordination ability of ligands that is reminiscent of the influence exerted by steric hindrance groups observed in biomolecules. Importantly, in the assembly of symmetrical ligands, only ligands modified with smaller spacer groups can form discrete supramolecular structures, underscoring the unique flexibility of dissymmetrical ligand assembly. It is anticipated that this approach is not limited to specific types of ligands and palladium metal but may also be broadly applicable to a variety of metal ions and organic ligands. Overall, this method provides an efficient means for constructing low-symmetry molecular cages, which are anticipated to have diverse applications in specialized molecular separation, chiral catalysis, sensing, and other related fields.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Ming Wang (mingwang358@jlu.edu.cn).

Materials availability

All materials generated in this study are available from the lead contact without restriction.

Data and code availability

Crystallographic data for the structure reported in this article have been deposited at the Cambridge Crystallographic Data Center under deposition number CCDC: 2263810. Copies of the data can be obtained free of charge from <https://www.ccdc.cam.ac.uk>. Videos S1, S2, S3, S4, and S5 are available in the supplemental information.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.xcrp.2023.101631>.

ACKNOWLEDGMENTS

We gratefully acknowledge support from the National Natural Science Foundation of China (22271116 and 22071079 for M.W.). We thank the staff at the BL17B1 beam-line of the National Facility for Protein Science in Shanghai (NFPS), Shanghai Advanced Research Institute, CAS, for providing technical support in X-ray diffraction data collection and analysis.

AUTHOR CONTRIBUTIONS

In this work, M.W. and H.Y. conceived and designed the experiments. H.Y. and Z.G. completed the synthesis. H.Y. completed NMR and crystal characterization. Q.B., Z.Z., and P.W. conducted ESI-MS characterization. M.W., H.Y., N.H., J.S., and X.J. analyzed the data and wrote the manuscript. All the authors discussed the results and commented on and proofread the final form of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: August 4, 2023

Revised: August 29, 2023

Accepted: September 18, 2023

Published: October 9, 2023

REFERENCES

- Dhara, A.K., and Nayak, A.K. (2022). Chapter 1 - Biological macromolecules: sources, properties, and functions. In *Biological Macromolecules*, A.K. Nayak, A.K. Dhara, and D. Pal, eds. (Academic Press), pp. 3–22.
- Seidel, S.R., and Stang, P.J. (2002). High-Symmetry Coordination Cages via Self-Assembly. *Acc. Chem. Res.* 35, 972–983. <https://doi.org/10.1021/ar010142d>.
- Naray-Szabo, G. (1982). In *Steric Effects in Biomolecules* (Elsevier Science Ltd).
- Wang, M., Wang, J., Zhou, P., Deng, J., Zhao, Y., Sun, Y., Yang, W., Wang, D., Li, Z., Hu, X., et al. (2018). Nanoribbons self-assembled from short peptides demonstrate the formation of polar zippers between β -sheets. *Nat. Commun.* 9, 5118. <https://doi.org/10.1038/s41467-018-07583-2>.
- Widlak, W. (2013). Protein Structure and Function. In *Molecular Biology: Not Only for Bioinformaticians*, W. Widlak, ed. (Springer Berlin Heidelberg), pp. 15–29.
- Pugh, G.C., Burns, J.R., and Howorka, S. (2018). Comparing proteins and nucleic acids for next-generation biomolecular engineering. *Nat. Rev. Chem.* 2, 113–130. <https://doi.org/10.1038/s41570-018-0015-9>.
- Görbitz, C.H., Törnroos, K.W., and Day, G.M. (2012). Single-crystal investigation of l-tryptophan with $Z' = 16$. *Acta Crystallogr. B* 68, 549–557. <https://doi.org/10.1107/S0108768112033484>.
- Kayushina, R., and KB, V. (1966). X-ray determination of structure of L-proline. *Sov. Phys. Crystallogr. USSR* 10, 698.
- Mostad, A., Nissen, H.M., and Romming, C. (1972). Crystal structure of L-tyrosine. *Acta Chem. Scand.* 26, 3819–3833.
- Northrop, B.H., Zheng, Y.-R., Chi, K.-W., and Stang, P.J. (2009). Self-Organization in Coordination-Driven Self-Assembly. *Acc. Chem. Res.* 42, 1554–1563. <https://doi.org/10.1021/ar900077c>.
- Saha, M.L., De, S., Pramanik, S., and Schmittel, M. (2013). Orthogonality in discrete self-assembly – survey of current concepts. *Chem. Soc. Rev.* 42, 6860–6909. <https://doi.org/10.1039/C3CS60098J>.
- Wang, W., Wang, Y.-X., and Yang, H.-B. (2016). Supramolecular transformations within discrete coordination-driven supramolecular architectures. *Chem. Soc. Rev.* 45, 2656–2693. <https://doi.org/10.1039/C5CS00301F>.
- Davies, J.A., Ronson, T.K., and Nitschke, J.R. (2022). Twisted rectangular subunits self-assemble into a ferritin-like capsule. *Chem* 8, 1099–1106. <https://doi.org/10.1016/j.chempr.2022.01.003>.
- Liu, W., and Stoddart, J.F. (2021). Emergent behavior in nanoconfined molecular containers. *Chem* 7, 919–947. <https://doi.org/10.1016/j.chempr.2021.02.016>.
- Xue, W., Ronson, T.K., Lu, Z., and Nitschke, J.R. (2022). Solvent Drives Switching between Δ and Λ Metal Center Stereochemistry of MgL_6 .

- Cubic Cages. *J. Am. Chem. Soc.* **144**, 6136–6142. <https://doi.org/10.1021/jacs.2c00245>.
16. Saha, S., Regeni, I., and Clever, G.H. (2018). Structure relationships between bis-monodentate ligands and coordination driven self-assemblies. *Coord. Chem. Rev.* **374**, 1–14. <https://doi.org/10.1016/j.ccr.2018.06.010>.
 17. Chakraborty, S., and Newkome, G.R. (2018). Terpyridine-based metallocupramolecular constructs: tailored monomers to precise 2D-motifs and 3D-metallocages. *Chem. Soc. Rev.* **47**, 3991–4016. <https://doi.org/10.1039/C8CS00030A>.
 18. Fielden, S.D.P., Leigh, D.A., and Woltering, S.L. (2017). Molecular Knots. *Angew. Chem., Int. Ed. Engl.* **56**, 11166–11194. <https://doi.org/10.1002/anie.201702531>.
 19. Zhang, Z., Li, Y., Song, B., Zhang, Y., Jiang, X., Wang, M., Tumbleton, R., Liu, C., Wang, P., Hao, X.-Q., et al. (2020). Intra- and intermolecular self-assembly of a 20-nm-wide supramolecular hexagonal grid. *Nat. Chem.* **12**, 468–474. <https://doi.org/10.1038/s41557-020-0454-z>.
 20. Zhang, H.-N., Feng, H.-J., Lin, Y.-J., and Jin, G.-X. (2023). Cation-Templated Assembly of 613 and 623 Metalla-Links. *J. Am. Chem. Soc.* **145**, 4746–4756. <https://doi.org/10.1021/jacs.2c13416>.
 21. Wu, K., Ronson, T.K., Goh, L., Xue, W., Heard, A.W., Su, P., Li, X., Vinković, M., and Nitschke, J.R. (2023). A Diverse Array of Large Capsules Transform in Response to Stimuli. *J. Am. Chem. Soc.* **145**, 11356–11363. <https://doi.org/10.1021/jacs.3c02491>.
 22. Liu, D., Li, K., Chen, M., Zhang, T., Li, Z., Yin, J.-F., He, L., Wang, J., Yin, P., Chan, Y.-T., and Wang, P. (2021). Russian-Doll-Like Molecular Cubes. *J. Am. Chem. Soc.* **143**, 2537–2544. <https://doi.org/10.1021/jacs.0c11703>.
 23. Wu, T., Jiang, Z., Bai, Q., Li, Y., Mao, S., Yu, H., Wojtas, L., Tang, Z., Chen, M., Zhang, Z., et al. (2021). Supramolecular triangular orthobipicupola: Self-assembly of a giant Johnson solid J₂₇. *Chem* **7**, 2429–2441. <https://doi.org/10.1016/j.chempr.2021.06.003>.
 24. Jiang, X., Yu, H., Shi, J., Bai, Q., Xu, Y., Zhang, Z., Hao, X.-Q., Li, B., Wang, P., Wu, L., and Wang, M. (2022). From Mechanically Interlocked Structures to Host–Guest Chemistry Based on Twisted Dimeric Architectures by Adjusting Space Constraints. *CCS Chem.* **4**, 2127–2139. <https://doi.org/10.31635/ccschem.021.202100948>.
 25. Lu, Y.-L., Zhang, X.-D., Qin, Y.-H., Song, J.-Q., Huang, Y.-H., Liu, C.-H., Chen, J.-J., Xu, H.-S., Pan, M., and Su, C.-Y. (2023). A robust protein-mimicking metallo-amine cage showing proton-driven allostery with water as the effector. *Chem* **9**, 2144–2160. <https://doi.org/10.1016/j.chempr.2023.03.019>.
 26. Siddique, R.G., Arachchige, K.S.A., Al-Fayaaad, H.A., Thoburn, J.D., McMurtrie, J.C., and Clegg, J.K. (2022). Controlling the Complexity and Interconversion Mechanisms in Self-Assembled [Fe₂L₃]⁴⁺ Helicates and [Fe₄L₄]⁸⁺ Cages. *Angew. Chem. Int. Ed.* **61**, e202115555. <https://doi.org/10.1002/anie.202115555>.
 27. Clegg, J.K., Li, F., and Lindoy, L.F. (2022). Oligo-β-diketones as versatile ligands for use in metallo-supramolecular chemistry: Recent progress and perspectives. *Coord. Chem. Rev.* **455**, 214355. <https://doi.org/10.1016/j.ccr.2021.214355>.
 28. Ronson, T.K., Carpenter, J.P., and Nitschke, J.R. (2022). Dynamic optimization of guest binding in a library of diastereomeric heteroleptic coordination cages. *Chem* **8**, 557–568. <https://doi.org/10.1016/j.chempr.2021.12.017>.
 29. Li, G., Ronson, T.K., Lavendomme, R., Huang, Z., Fuertes-Espinosa, C., Zhang, D., and Nitschke, J.R. (2023). Enantiopure FeL₄L₄ cages bind steroids stereoselectively. *Chem* **9**, 1549–1561. <https://doi.org/10.1016/j.chempr.2023.03.011>.
 30. Harris, K., Fujita, D., and Fujita, M. (2013). Giant hollow M_nL_{2n} spherical complexes: structure, functionalisation and applications. *Chem. Commun.* **49**, 6703–6712. <https://doi.org/10.1039/C3CC43191F>.
 31. Lewis, J.E.M. (2022). Molecular engineering of confined space in metal–organic cages. *Chem. Commun.* **58**, 13873–13886. <https://doi.org/10.1039/D2CC05560K>.
 32. Han, M., Engelhard, D.M., and Clever, G.H. (2014). Self-assembled coordination cages based on banana-shaped ligands. *Chem. Soc. Rev.* **43**, 1848–1860. <https://doi.org/10.1039/C3CS60473J>.
 33. Findlay, J.A., Patil, K.M., Gardiner, M.G., MacDermott-Opeskin, H.I., O'Mara, M.L., Kruger, P.E., and Preston, D. (2022). Heteroleptic Tripalladium(II) Cages. *Chem. Asian J.* **17**, e202200093. <https://doi.org/10.1002/asia.202200093>.
 34. Chu, D.I., Balmert, L.C., Chen, L., Arkin, C., Meyer, T., Rosoklija, I., Bowen, D.K., Hodgkins, K.S., Bowman, R.M., Cheng, E.Y., et al. (2021). Boosting Enantioselectivity of Chiral Molecular Catalysts with Supramolecular Metal–Organic Cages. *J. Urol.* **205**, 1180–1188. <https://doi.org/10.31635/ccschem.021.202100847>.
 35. Li, C., Wang, Y., Lu, Y., Guo, J., Zhu, C., He, H., Duan, X., Pan, M., and Su, C. (2020). An iridium(III)–palladium(II) metal–organic cage for efficient mitochondria-targeted photodynamic therapy. *Chin. Chem. Lett.* **31**, 1183–1187. <https://doi.org/10.1016/j.cclet.2019.09.034>.
 36. Tateishi, T., Yoshimura, M., Tokuda, S., Matsuda, F., Fujita, D., and Furukawa, S. (2022). Coordination/metal–organic cages inside out. *Coord. Chem. Rev.* **467**, 214612. <https://doi.org/10.1016/j.ccr.2022.214612>.
 37. Lee, H., Tessarolo, J., Langbehn, D., Baksi, A., Herges, R., and Clever, G.H. (2022). Light-Powered Dissipative Assembly of Diazocine Coordination Cages. *J. Am. Chem. Soc.* **144**, 3099–3105. <https://doi.org/10.1021/jacs.1c12011>.
 38. Tang, X., Jiang, H., Si, Y., Rampal, N., Gong, W., Cheng, C., Kang, X., Fairen-Jimenez, D., Cui, Y., and Liu, Y. (2021). Endohedral functionalization of chiral metal–organic cages for encapsulating achiral dyes to induce circularly polarized luminescence. *Chem* **7**, 2771–2786. <https://doi.org/10.1016/j.chempr.2021.07.017>.
 39. Tessarolo, J., Lee, H., Sakuda, E., Umakoshi, K., and Clever, G.H. (2021). Integrative Assembly of Heteroleptic Tetrahedra Controlled by Backbone Steric Bulk. *J. Am. Chem. Soc.* **143**, 6339–6344. <https://doi.org/10.1021/jacs.1c01931>.
 40. Bobylev, E.O., Zeng, Y., Weijertse, K., Koelman, E., Meijer, E.M., de Bruin, B., Kros, A., and Reek, J.N. (2023). The application of M₁₂L₂₄ nanocages as cell-specific siRNA delivery agents in vitro. *Chem* **9**, 1578–1593. <https://doi.org/10.1016/j.chempr.2023.03.018>.
 41. Fujita, M., Yazaki, J., and Ogura, K. (1990). Preparation of a macrocyclic polynuclear complex, [(en)Pd(4,4'-bpy)]₄(NO₃)₈ (en = ethylenediamine, bpy = bipyridine), which recognizes an organic molecule in aqueous media. *J. Am. Chem. Soc.* **112**, 5645–5647. <https://doi.org/10.1021/ja00170a042>.
 42. Bunzen, J., Iwasa, J., Bonakdarzadeh, P., Numata, E., Rissanen, K., Sato, S., and Fujita, M. (2012). Self-Assembly of M₂₄L₄₈ Polyhedra Based on Empirical Prediction. *Angew. Chem. Int. Ed.* **51**, 3161–3163. <https://doi.org/10.1002/anie.201108731>.
 43. Fujita, D., Ueda, Y., Sato, S., Yokoyama, H., Mizuno, N., Kumasaka, T., and Fujita, M. (2016). Self-Assembly of M₃₀L₆₀ Icosidodecahedron. *Chem* **1**, 91–101. <https://doi.org/10.1016/j.chempr.2016.06.007>.
 44. Platzek, A., Juber, S., Yurtseven, C., Hasegawa, S., Schneider, L., Drechsler, C., Ebbert, K.E., Rudolf, R., Yan, Q.-Q., Holstein, J.J., et al. (2022). Endohedrally Functionalized Heteroleptic Coordination Cages for Phosphate Ester Binding. *Angew. Chem. Int. Ed.* **61**, e202209305. <https://doi.org/10.1002/anie.202209305>.
 45. Schulte, T.R., Holstein, J.J., and Clever, G.H. (2019). Chiral Self-Discrimination and Guest Recognition in Helicene-Based Coordination Cages. *Angew. Chem. Int. Ed.* **58**, 5562–5566. <https://doi.org/10.1002/anie.201812926>.
 46. Preston, D., Sutton, J.J., Gordon, K.C., and Crowley, J.D. (2018). A Nona-nuclear Heterometallic Pd₃Pt₆ “Donut”-Shaped Cage: Molecular Recognition and Photocatalysis. *Angew. Chem. Int. Ed.* **57**, 8659–8663. <https://doi.org/10.1002/anie.201804745>.
 47. Lisboa, L.S., Findlay, J.A., Wright, L.J., Hartinger, C.G., and Crowley, J.D. (2020). A Reduced-Symmetry Heterobimetallic [PdPtL₄]⁴⁺ Cage: Assembly, Guest Binding, and Stimulus-Induced Switching. *Angew. Chem. Int. Ed.* **59**, 11101–11107. <https://doi.org/10.1002/anie.202003220>.
 48. Ueda, M., Kishida, N., Catti, L., and Yoshizawa, M. (2022). Caged bulky organic dyes in a polyaromatic framework and their spectroscopic peculiarities. *Chem. Sci.* **13**, 8642–8648. <https://doi.org/10.1039/D2SC02308C>.
 49. Kishida, N., Matsumoto, K., Tanaka, Y., Akita, M., Sakurai, H., and Yoshizawa, M. (2020). Anisotropic Contraction of a Polyaromatic Capsule and Its Cavity-Induced Compression Effect. *J. Am. Chem. Soc.* **142**, 9599–9603. <https://doi.org/10.1021/jacs.0c02932>.
 50. Li, R.-J., Tarzia, A., Posligua, V., Jelfs, K.E., Sanchez, N., Marcus, A., Baksi, A., Clever, G.H., Fadaei-Tirani, F., and Severin, K. (2022). Orientational self-sorting in cuboctahedral Pd

- cages. *Chem. Sci.* 13, 11912–11917. <https://doi.org/10.1039/D2SC03856K>.
51. Sudan, S., Li, R.-J., Jansze, S.M., Platzeck, A., Rudolf, R., Clever, G.H., Fadaei-Tirani, F., Scopelliti, R., and Severin, K. (2021). Identification of a Heteroleptic $\text{Pd}_6\text{L}_6\text{L}'_6$ Coordination Cage by Screening of a Virtual Combinatorial Library. *J. Am. Chem. Soc.* 143, 1773–1778. <https://doi.org/10.1021/jacs.0c12793>.
52. Sato, S., Iida, J., Suzuki, K., Kawano, M., Ozeki, T., and Fujita, M. (2006). Fluorous Nanodroplets Structurally Confined in an Organopalladium Sphere. *Science* 313, 1273–1276. <https://doi.org/10.1126/science.1129830>.
53. Saha, S., Holzapfel, B., Chen, Y.-T., Terlinden, K., Lill, P., Gatsogiannis, C., Rehage, H., and Clever, G.H. (2018). Rational Design of an Amphiphilic Coordination Cage-Based Emulsifier. *J. Am. Chem. Soc.* 140, 17384–17388. <https://doi.org/10.1021/jacs.8b10991>.
54. Johnson, A.M., and Hooley, R.J. (2011). Steric Effects Control Self-Sorting in Self-Assembled Clusters. *Inorg. Chem.* 50, 4671–4673. <https://doi.org/10.1021/ic2001688>.
55. Preston, D., Barnsley, J.E., Gordon, K.C., and Crowley, J.D. (2016). Controlled Formation of Heteroleptic $[\text{Pd}_2(\text{L})_2(\text{Lb})_2]^{4+}$ Cages. *J. Am. Chem. Soc.* 138, 10578–10585. <https://doi.org/10.1021/jacs.6b05629>.
56. Elliott, A.B.S., Lewis, J.E.M., van der Salm, H., McAdam, C.J., Crowley, J.D., and Gordon, K.C. (2016). Luminescent Cages: Pendant Emissive Units on $[\text{Pd}_2\text{L}_4]^{4+}$ “Click” Cages. *Inorg. Chem.* 55, 3440–3447. <https://doi.org/10.1021/acs.inorgchem.5b02843>.
57. Liu, Y., Liao, S.-H., Dai, W.-T., Bai, Q., Lu, S., Wang, H., Li, X., Zhang, Z., Wang, P., Lu, W., and Zhang, Q. (2023). Controlled Construction of Heteroleptic $[\text{Pd}_2(\text{L})_2(\text{Lb})(\text{LC})]^{4+}$ Cages: A Facile Approach for Site-Selective endo-Functionalization of Supramolecular Cavities. *Angew. Chem. Int. Ed.* 62, e202217215. <https://doi.org/10.1002/anie.202217215>.
58. Yan, X., Wei, P., Liu, Y., Wang, M., Chen, C., Zhao, J., Li, G., Saha, M.L., Zhou, Z., An, Z., et al. (2019). Endo- and Exo-Functionalized Tetraphenylethylene $\text{M}_{12}\text{L}_{24}$ Nanospheres: Fluorescence Emission inside a Confined Space. *J. Am. Chem. Soc.* 141, 9673–9679. <https://doi.org/10.1021/jacs.9b03885>.
59. Johnson, A.M., Moshe, O., Gamboa, A.S., Langloss, B.W., Limtiaco, J.F.K., Larive, C.K., and Hooley, R.J. (2011). Synthesis and Properties of Metal–Ligand Complexes with Endohedral Amine Functionality. *Inorg. Chem.* 50, 9430–9442. <https://doi.org/10.1021/ic201092s>.
60. Freye, S., Michel, R., Stalke, D., Pawliczek, M., Frauendorf, H., and Clever, G.H. (2013). Template Control over Dimerization and Guest Selectivity of Interpenetrated Coordination Cages. *J. Am. Chem. Soc.* 135, 8476–8479. <https://doi.org/10.1021/ja403184a>.
61. Krick, M., Holstein, J., Würtele, C., and Clever, G.H. (2016). Endohedral dynamics of push–pull rotor-functionalized cages. *Chem. Commun.* 52, 10411–10414. <https://doi.org/10.1039/C6CC04155H>.
62. Löffler, S., Lübken, J., Wuttke, A., Mata, R.A., John, M., Dittrich, B., and Clever, G.H. (2016). Internal dynamics and guest binding of a sterically overcrowded host. *Chem. Sci.* 7, 4676–4684. <https://doi.org/10.1039/C6SC00985A>.
63. Lewis, J.E.M., Elliott, A.B.S., McAdam, C.J., Gordon, K.C., and Crowley, J.D. (2014). ‘Click’ to functionalise: synthesis, characterisation and enhancement of the physical properties of a series of exo- and endo-functionalised Pd_2L_4 nanocages. *Chem. Sci.* 5, 1833–1843. <https://doi.org/10.1039/C4SC00434E>.
64. Yu, H., Li, J., Shan, C., Lu, T., Jiang, X., Shi, J., Wojtas, L., Zhang, H., and Wang, M. (2021). Conformational Control of a Metallo-Supramolecular Cage via the Dissymmetrical Modulation of Ligands. *Angew. Chem. Int. Ed.* 60, 26523–26527. <https://doi.org/10.1002/anie.202111430>.
65. Lewis, J.E.M. (2022). Pseudo-heterolepticity in Low-Symmetry Metal–Organic Cages. *Angew. Chem. Int. Ed.* 61, e202212392. <https://doi.org/10.1002/anie.202212392>.
66. Gao, X., Fu, A.-Y., and Wang, Y.-Y. (2016). Shape-Asymmetry Supramolecular Isomerism in Asymmetrical Ligand PCs and the Expression Method of Three-Level Isomerism. *Inorg. Chem.* 55, 4330–4334. <https://doi.org/10.1021/acs.inorgchem.6b00081>.
67. Li, R.-J., Marcus, A., Fadaei-Tirani, F., and Severin, K. (2021). Orientational self-sorting: formation of structurally defined Pd_4L_8 and Pd_6L_{12} cages from low-symmetry dipyrrolic ligands. *Chem. Commun.* 57, 10023–10026. <https://doi.org/10.1039/D1CC03828A>.
68. Klein, C., Gütz, C., Bogner, M., Topić, F., Rissanen, K., and Lützen, A. (2014). A New Structural Motif for an Enantiomerically Pure Metallosupramolecular Pd_4L_8 Aggregate by Anion Templating. *Angew. Chem. Int. Ed.* 53, 3739–3742. <https://doi.org/10.1002/anie.201400626>.
69. Käseborn, M., Holstein, J.J., Clever, G.H., and Lützen, A. (2018). A Rotaxane-like Cage-in-Ring Structural Motif for a Metallosupramolecular Pd_6L_{12} Aggregate. *Angew. Chem. Int. Ed.* 57, 12171–12175. <https://doi.org/10.1002/anie.201806814>.
70. Pappas, C.G., Mandal, P.K., Liu, B., Kauffmann, B., Miao, X., Komáromy, D., Hoffmann, W., Manz, C., Chang, R., Liu, K., et al. (2020). Emergence of low-symmetry foldamers from single monomers. *Nat. Chem.* 12, 1180–1186. <https://doi.org/10.1038/s41557-020-00565-2>.